

Don't Believe the AI Hype

May 21, 2024 | **DARON ACEMOGLU**

BOSTON – According to tech leaders and many pundits and academics, artificial intelligence is poised to transform the world as we know it through unprecedented productivity gains. While some believe that machines soon will do everything humans can do, ushering in a new age of boundless prosperity, other predictions are at least more grounded. For example, Goldman Sachs predicts that generative AI will boost global GDP by 7% over the next decade, and the McKinsey Global Institute anticipates that the annual GDP growth rate could increase by 3-4 percentage points between now and 2040. For its part, *The Economist* expects that AI will create a blue-collar bonanza.

Is this realistic? As I note in a recent paper, the outlook is far more uncertain than most forecasts and guesstimates suggest. Still, while it is basically impossible to predict with any confidence what AI will do in 20 or 30 years, one can say something about the next decade, because most of these near-term economic effects must involve existing technologies and improvements to them.

It is reasonable to suppose that AI's biggest impact will come from automating some tasks and making some workers in some occupations more productive. Economic theory provides some guidance for assessing these aggregate effects. According to Hulten's theorem (named for economist Charles Hulten), aggregate "total factor productivity" (TFP) effects are simply the product of the share of tasks that are automated multiplied by the average cost savings.

While average cost savings are difficult to estimate and will vary by activity, there have already been some careful studies of AI's effects on certain tasks. For example, Shakked Noy and Whitney Zhang have examined the impact of ChatGPT on simple writing tasks (such as summarizing documents or writing routine grant proposals or marketing material), while Erik Brynjolfsson, Danielle Li, and Lindsey Raymond have assessed the use of AI assistants in customer service. Taken together, this research suggests that currently available generative-AI tools yield average labor-cost savings of 27% and overall cost savings of 14.4%.

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What about the share of tasks that will be affected by AI and related technologies? Using numbers from recent studies, I estimate this to be around 4.6%, implying that AI will increase TFP by only